Exploring Biology in Society

Content Standards:

Simple Living Organisms in the Environment

Learning Indicators:

Life Processes And Economic Importance Of Some Common Simple Living Organisms (Rhizopus, Moss And Fern).

Rhizopus

Scientific Classification:

Domian: Eukarya

Kingdom: Fungi

Phylum: Zygomycota

Genus: *Rhizopus*

Rhizopus stolonifer is also known as black bread mold.

Adaptive features of *Rhizopus stolonifer*

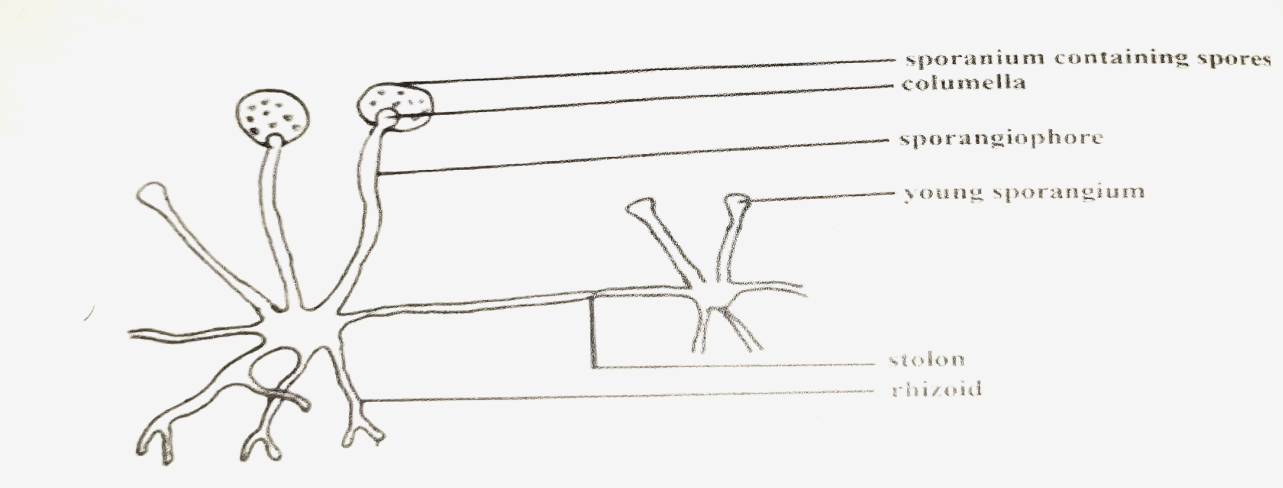
• Presence of sporangia produces spores

• Presence of sporangiophore holds sporangia in position for dispersal of spores

• Mycelium for vegetative growth

• Spores for asexual reproduction

• Rhizoids for absorption of nutrients and anchorage



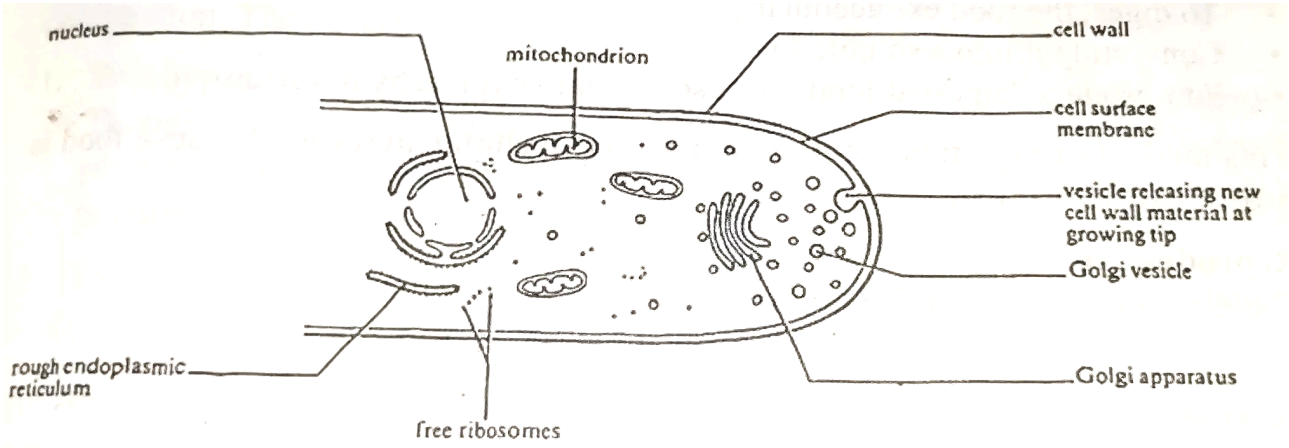
Rhizopus forms a mass of soft, white silky branching threads, called **hypae** (singular-hypha). A mass of hyphae is called **mycellium**. The hyphae look like a thin layer of cotton wool on the surface of bread. The hypha is cylindrical or tube-like. The wall of the hypha consists of chitin which is made up of proteins and carbohydrates. The chitin is lined by a thin layer of cytoplasm.

Several nuclei lie inside the cytoplasm. The centre of the tube of hypha is occupied by the vacuole. There is no cross-wall in a hypha which is less than three days old to divide it into separate cells. Such hyphae are described as **non - septate**. Old hyphae develop cross – walls and are described as **septate hyphae**. The mycelium consists of **three types of hypha**:

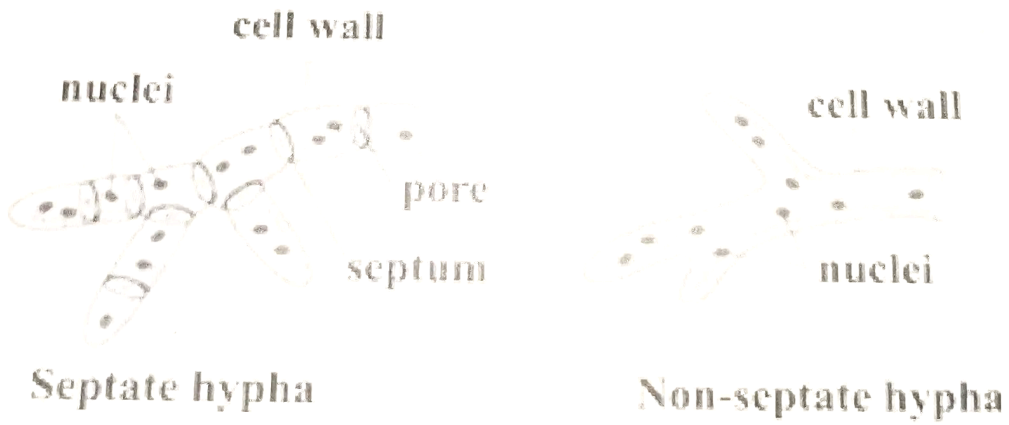
1. Stolons: These grow horizontally on the substrate.

2. Rhizoids: They are root-like. They attach and penetrate into food at points where stolons make contact with the substrate. Rhizoids absorb food substances saprophytically.

3. Sporangiophores: These are the hyphae that grow vertically upwards. Their tips bear round reproductive structures called sporangia



Drawing of L S hypha under electron microscope



Functions of parts of Rhizopus

SN	Part	Function
1	Hyphae	Spread the fungus and absorb nutrients
2	Mycelium	Spread the fungus
3	Sporangium	Produces the spores
4	Sporangiophore	Holds up the sporangium
5	Stolon	Allows fungi to spread
6	Columella	Spore release

Method of Feeding in Rhizopus

• *Rhizopus* undergoes saprophytic mode of nutrition;

• The hyphae penetrate into the substrate/decaying organic matter;

• And secrete digestive enzymes on the food;

• To digest the food extracellularly;

• Converting it into a soluble form;

• End-products/digested food are absorbed into hyphae by diffusion

This method of digestion is referred to as **extracellular digestion** because food is broken down externally.

Reproduction

There are two methods of reproduction in *Rhizopus*:

Asexual reproduction

Sexual reproduction

Asexual reproduction: this method of reproduction occurs when conditions are favourable. During asexual reproduction, a number of sporangiophores arise from the substrate. At the tip of each of these hyphae, a spherical swelling, called a sporangium (sporangia) is formed. The sporangia are seen as little black dots on the substrate. The sporangium contains cytoplasm and numerous nuclei. In a later development, the cytoplasm with its nuclei moves to the periphery of the sporangium. The central portion of the sporangium, containing vacuoles, called columella becomes cut off from the rest of the hypha. The protoplasm of the sporangium, now concentrated in the periphery, breaks up into a number of groups. Each group contains about three or four nuclei. Each of these little bodies formed acquires a wall, to become a spore. Both the sporangium and columella absorb water and swell. The columella then bulges strongly into the sporangium to cause a great pressure to develop inside it. The pressure burst the sporangium wall, liberating the spores. Any spore that falls on a suitable medium like a bit of damp bread germinates. New hyphae are sent out which branch severally and eventually produce new mycelium.

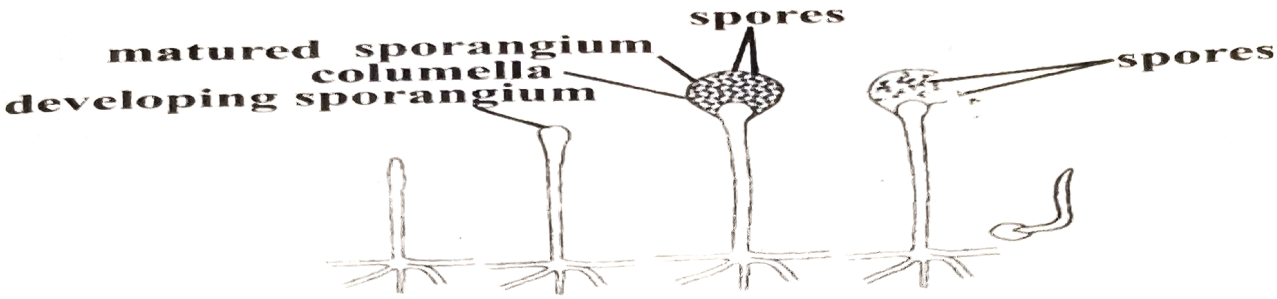
After germination, growth is favoured by:

Suitable temperature

Presence of sufficient nutrients

Presence of moisture in the substrate and in the environment

Food substrate must be easily digestible



Asexual reproduction in *Rhizopus*

Sexual reproduction: the process of sexual reproduction in *Rhizopus* is called conjugation. The processes are as follows:

Two hyphae deep in the mycelium approach each other with their tips or ends. Though all hyphae are structurally alike, they are really of two kinds called positive (+) and negative (-) strains. Sexual process can occur between a positive (+) strain and a (-) strain only.

The hyphae meet and their ends swell with cytoplasm forming progametangia (singular: progametangium).

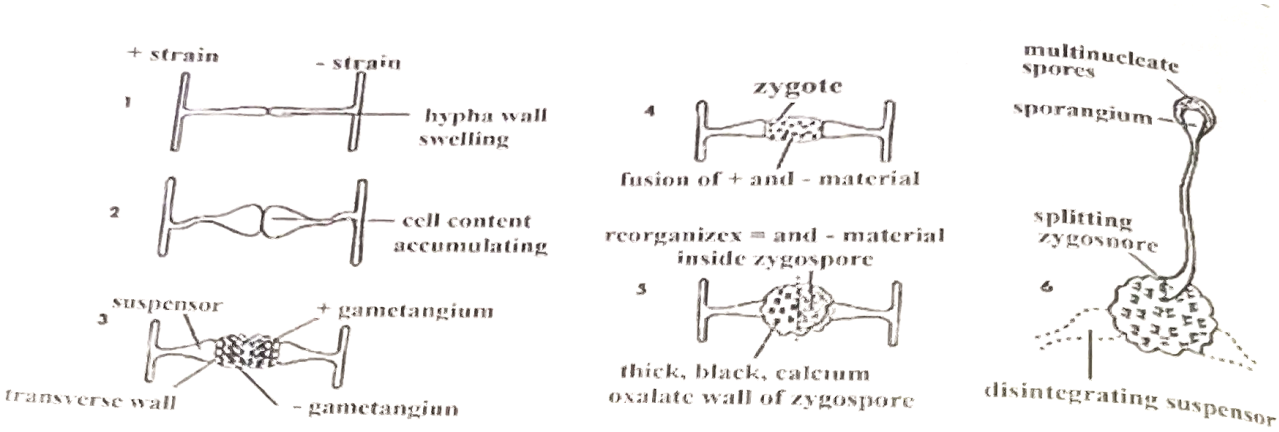
Soon after meeting, the apex of each hypha is cut off by a transverse wall. The isolated front ends of each branched hypha are called a gametangium. The remaining branch is called a suspensor.

Next the end walls separating the contents of the two gametangia disappear. Their protoplasm mix and the nuclei fuse in pairs. Each pair consists of one nucleus from one gametangium.

The walls of the gametangia become thickened, rounded, black and its whole surface covered with round-shaped swellings.

The structure formed is called zygospore. The zygospore can resist dehydration and other unfavorable conditions such as excessive heat. It can be carried by the air. The zygospore germinates to produce aerial hyphae with sporangium containing spores.

The processes are illustrated below:



Conjugation in *Rhizopus*

Conjugation in *Rhizopus*

Organisms that reproduce by conjugation:

Paramecium sp. *Zygnema* sp. *Spirogyra* sp. *Rhizopus*/*Mucor* sp. *Chlamydomonas* sp.

Facts about bread mould:

Bread mould consists of threadlike structures called Hyphae

Hyphae are tubular with no cross walls and are multinucleate. Each nucleus is haploid.

Large numbers of hyphae are called a mycelium

The hyphae digest the substrate on which they grows

Rhizoids provide extra surface area for absorption of the digested material

Adaptive features of spores for dispersal

It is light or small in size to be carried by wind

the spores are numerous to ensure survival

The thickened wall of the spores protect it from unfavourable environment

Factors that enhance growth of *Rhizopus*

• Moisture to activate enzymes to digest food

• Suitable temperature for enzymatic activity between 15°C and 30°C

• Presence of dead organic matter/nutrients for growth of mycelia

Ways of preventing mould growth

• By keeping bread/substrate in dry place

• keep substrate /bread in tightly sealed polythene bag

• store substrate/bread in freezer/fridge

Economic importance of *Rhizopus* / Bread mould

• Spoilage of food/causes food poisoning;

• Recycling of nutrients;

• Decay of organic materials.

Effect of *Rhizopus* growth on substrate, i.e bread

• Changes colour

• Spoilage

• Reduces economic value

• Contamination

• And reduces nutrient value

Effect of freezing on *Rhizopus*

• Keep food at low temperature fridge to prevent growth and multiplication of *Rhizopus*

• Low temperature kills *Rhizopus*

• Freezing deactivate digestive enzymes of *Rhizopus* making it unable to decompose the food

Why bread/ kenkey or substrate becomes mouldy when left at room temperature for days

• Mould spores are borne in the air,

• Bread provides a suitable substrate for mould spores,

• Spores land on bread and begin to multiply

• Under optimum temperature and pressure

SCIENTIFIC INQUIRY

Culturing *Rhizopus*

Rhizopus grows better at temperatures between 15°C - 30°C. The method of culturing include:

Obtain a piece of bread Moisten it with water

Place in a petri dish and cover

Keep in a cupboard for three days

Use forceps to obtain mycelia

Mount under low power microscope

Turn to high power

Preparing a temporary slide of *Rhizopus*

Materials

Microscope slide

Cover slip

Mounted needle/large pin

Forceps

Procedure

Take a clean and transparent glass slide

Use mounted needle or forceps to transfer a thin specimen onto the slide

Add a drop of methylene blue (a stain)

With the help of the finger and a mounted needle, lower the clean cover slip gently to cover the specimen

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***Rhizopus* mounted under the light microscope**

Observations

The black represent matured sporangia and the white represent immatured sporangia

Mutualistic association between alga and fungus to form lichen

Lichen is made up of a fungus and a unicellular alga living together. The fungus protects the alga and absorbs water and mineral salt from the surroundings. The algae, makes food by photosynthesis and in turn, provide the fungus with nutrients and oxygen.

Differences between specimens

Spirogyra and Bread mould

SN	***Spirogyra***	**Bread mould**
1	Chlorophyll/ green colour present	Chlorophyll/ green colour absent;
2	Mycelium/ hyphae absent	mycelium/ hyphae present;
3	Spores absent	Spores present;
4	Mucilage/slimy body present	Mucilage/slimy body absent;
5	Sporangiophore absent	sporangiophore present;
6	Sporangium absent	sporangium present;
7	Filamentous	Non-filamentous;
8	Septate	Non-septate

Differences between *Rhizopus* and bracket fungus

Rhizopus	Bracket fungus
Relatively smaller in size	Relatively larger in size
Reproduce asexually by sporangia	Asexual reproduction is absent
Non septate hypha	Septate hypha

Mosses

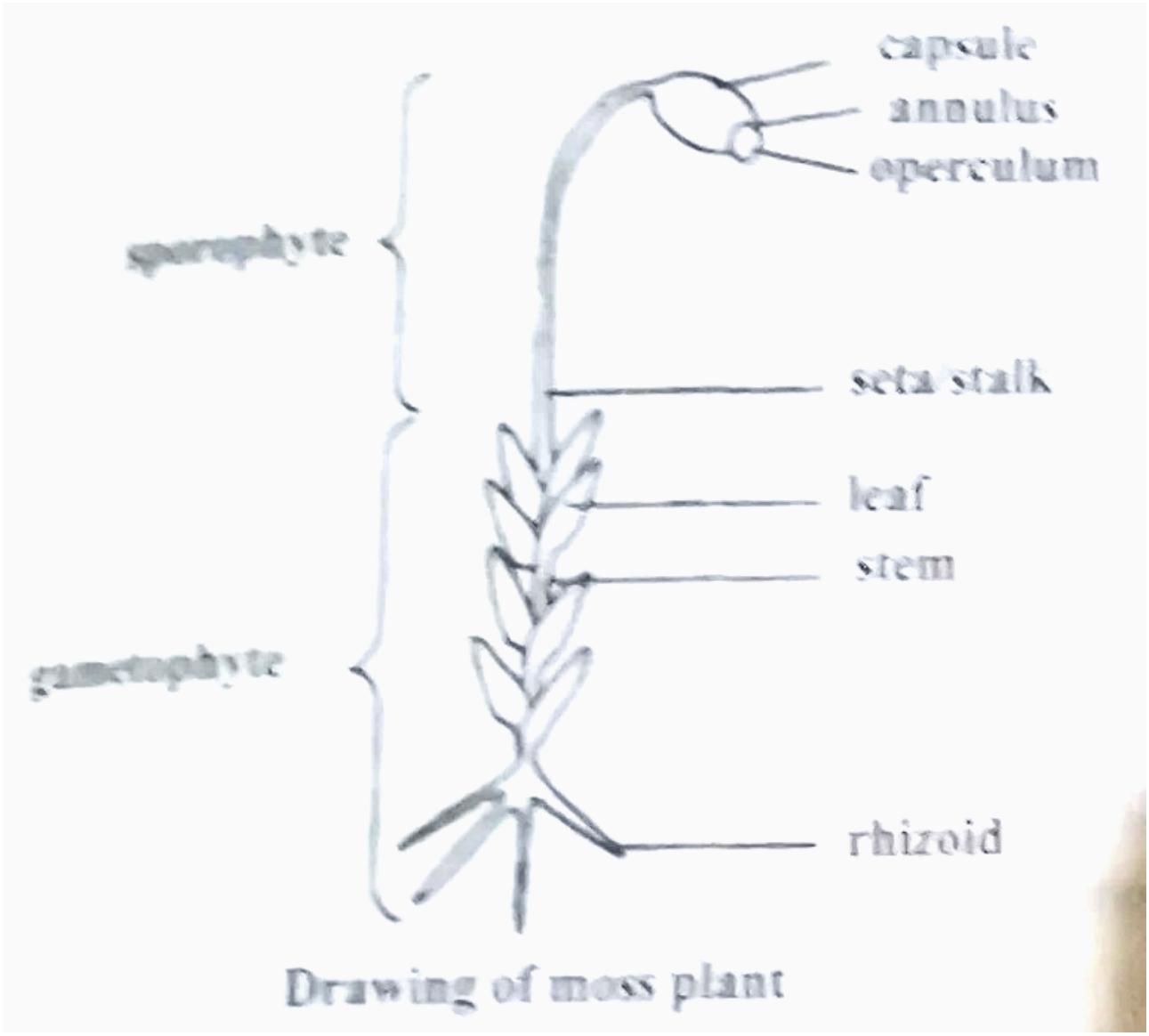
Scientific Classification:

Domain: Eukarya

Kingdom: Plantae

Phylum: Bryophyta

Genus: *Musci*



capsule

annulus

operculum

sporophyte

seta/stalk

leaf

stem

gametophyte

rhizoid

Drawing of moss plant

Habitat: Damp or moist terrestrial areas. Also grows on branches of rain forest trees

Anatomy

At first glance, one moss plant is seen to have structures that look like stem, leaves and roots. Microscopic examination shows that these structures are not true roots, leaves and stems, as found in higher plants. The stem has no conducting tissue; i.e. is phloem and xylem. The leaves are only one cell thick except in the centre. The rhizoids consist of only a few cells thick. A typical tufted moss bears a cluster of rhizoids at its base. The principal role of the rhizoid is for anchorage. Together with other parts, the rhizoid can absorb water and mineral salts. The leaf-like structures are arranged spirally along the stem. The stem, leaves and the rhizoid constitute the **gametophyte**. At a stage in the life of the moss, the gametophyte produces a stalk, called **seta**, with a capsule having a cap called the calyptra (a remnant of the archegonium) at its tip. This is the **sporophyte**. The sporophyte is attached to the gametophyte by a foot through which water and nutrients are absorbed. However, the sporophyte is not totally dependent on the gametophyte. The sporophyte is semi-parasitic on the gametophyte. ***Funaria*** and ***Brachymenium*** are examples of tufted mosses.

Mode of life

Nutrition: Mosses are holophytic. They are able to manufacture their own food by photosynthesis. Water, mineral salts and carbon dioxide are absorbed through all parts of the body surface.

Reproduction: is by sexual and asexual means which alternate. The moss therefore shows **alternation of generations**. The moss has a male sex organ called **antheridium** (plural:antheridia) and a female sex organ called **archegonium** (plural: archegonia). These reproductive structures are produced at the tips of the male and female gametophyte. The antheridium produces unicellular haploid male gametes called **antherozoids**. The archegonium produces **oosphere or ovum or egg**. The antherozoid has flagellum and can, therefore swim in a film of water covering the surface of the plant. The bi-flagellated antherozoids enters the archegonium and fuses with the ovum. The fertilized ovum in the archegonium develops into the sporophyte, which consists of a long stalk, called seta, bearing a capsule. The capsule bursts to

release the spores, which are dispersed by air currents. In suitable conditions, the spores germinate to produce a branching green filamentous structure called the **protonema**. A new gametophyte plant grows from the protonema. The life cycle of the moss plant as described above is termed as **alternation of generations**. The haploid gametophyte (producing gametes for sexual reproduction) alternates with diploid sporophyte (producing spores for asexual reproduction). **Production of large number of spores is for increasing the chances of survival and is an adaptation to land conditions.** In mosses, the gametophyte is the **dominant generation**.

antherozoid

oosphere

antheridium

archegonium

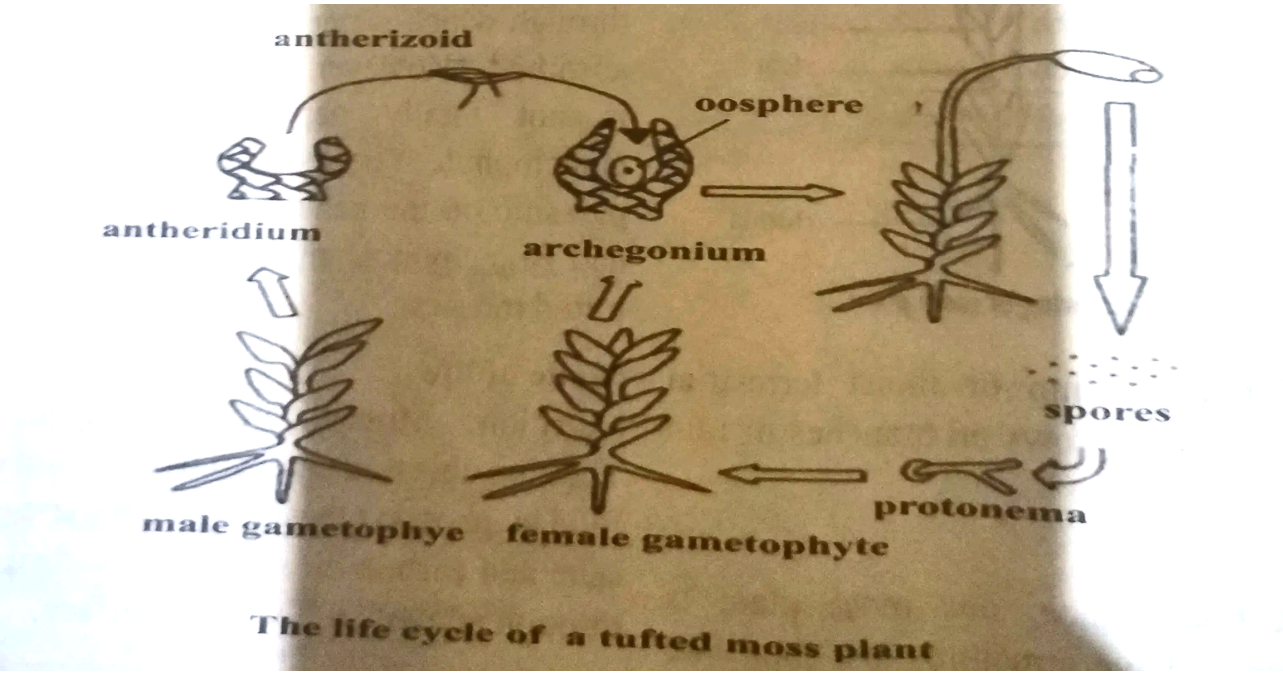
male gametophyte

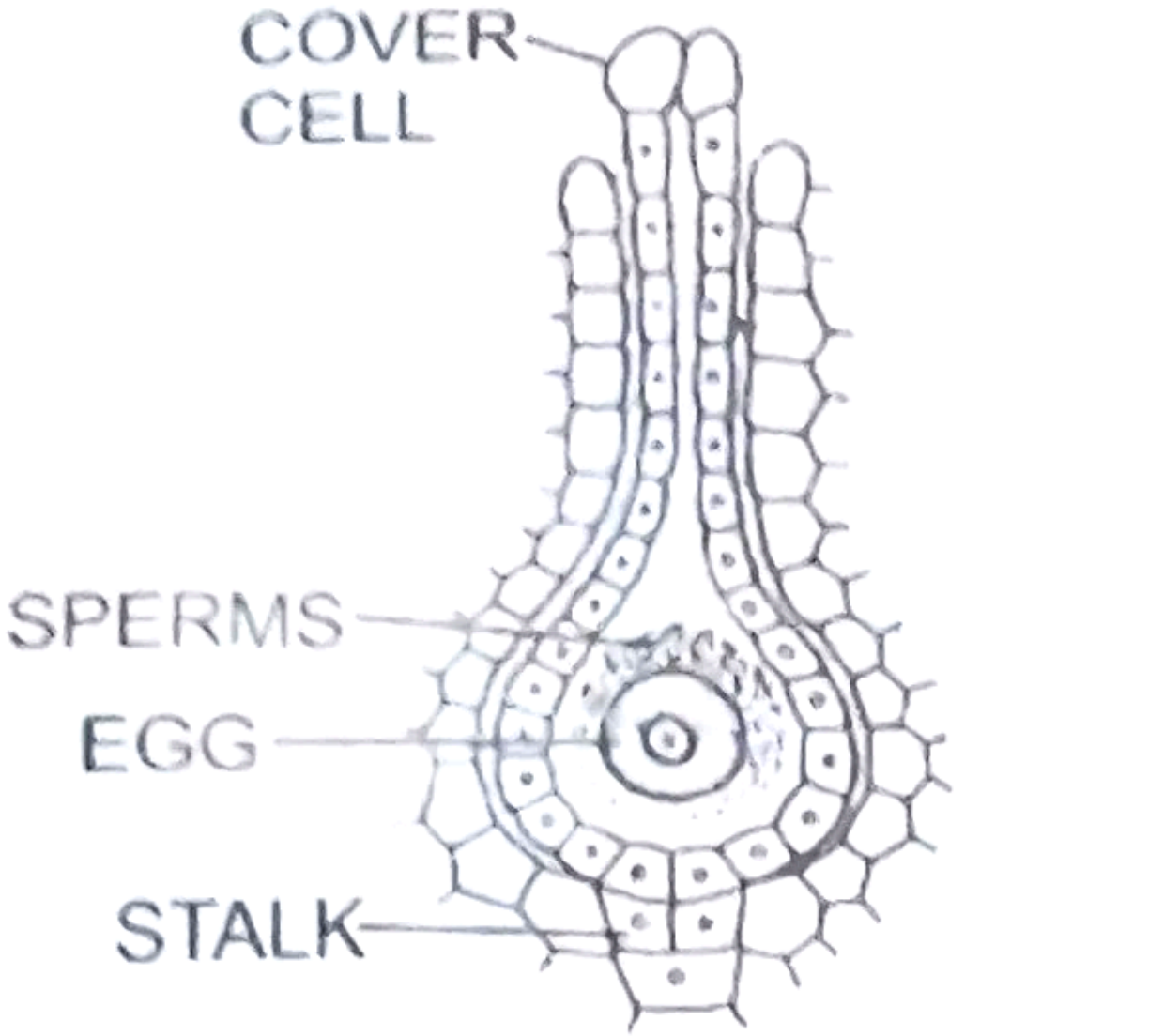
female gametophyte

spores

protonema

The life cycle of a tufted moss plant





A mature archegonium

Summary of the life cycle of a moss

• The green moss plant (haploid generation) has all its cells only having one single set of chromosomes. The plant at this stage is also called **gametophyte**.

• The gametophyte now forms male sex organ (antheridium) and female sex organ (archegonium) in which the gametes develop. The archegonium produces the egg cells (oosphers) while antheridium forms the antherozoids.

• The bi-flagellated antherozoids swim through a film of water to reach the archegonium where it fuses with the egg cell to form a zygote.

• The zygote develops in the archegonium into the diploid generation, the sporophyte which normally consists of a capsule on a stem.

• In the capsule, spores develop but are released when the capsule bursts during maturity.

• When a spore falls on a damp place, it germinates into protonema (mesh of cell filaments).

• The protonema builds some cusps from which the new leafy moss plant arise.

Adaptive features of the moss

• Leaves are small in size and spirally arranged to reduce the rate of transpiration

• Leaves have chlorophyll to undergo photosynthesis

• It has multicellular rhizoids for anchorage and absorption of water and mineral salts

• Has long stalk of sporophyte which exposes the capsule for easy dispersal of spores.

Importance of moss

• Check erosion

• The peat deposits of *Sphagnum* (an aquatic moss) are cut into blocks, dried and used as fuel

• Used as soil conditioners

• Producers in food chain/ source of food (peat moss)

• Improve moisture content in the formation of roots during marcotting

• Source of oxygen through photosynthesis

• Early colonizers in succession

• Peat moss due to its ability to retain water is used for packing plants and cut flowers which are transported to far away places

NB: ***Sphagnum*** is a bryophyte, commonly called as ***bog moss or peat moss***. It is ***hygroscopic*** and possesses a remarkable water holding capacity. Hence, it is used as a ***packing material*** in the transportation of flowers, live plants, tubers, bulbs, seedlings, etc.

NB: ***Metagenesis*** in an organism refers to the reproduction characterised by the alteration of a sexual generation and a generation that reproduces asexually, i.e. alteration of generations.

Uses of Mosses in Environmental Monitoring

Mosses play a significant role in environmental monitoring due to their sensitivity to environmental changes and their ability to accumulate pollutants. Here are some ways mosses are used in environmental monitoring:

> Air Quality Monitoring: Mosses are sensitive to air pollutants, such as heavy metals, sulfur dioxide, and nitrogen oxides. By analyzing the chemical composition of mosses, researchers can assess the levels of air pollution in a given area.
>
> Heavy Metal Monitoring: Mosses can absorb and accumulate heavy metals, such as lead, cadmium, and mercury, from the air and soil. By analyzing the heavy metal content of mosses, researchers can identify areas with high levels of heavy metal pollution.

Water Quality Monitoring: Mosses can be used to monitor water quality by detecting changes in water chemistry and detecting the presence of pollutants.

Biomonitoring: Mosses can serve as biomarkers for environmental pollution, providing an early warning system for detecting changes in environmental conditions.

Ferns (Pteridophytes)

Ferns are the most diverse group of seedless plants. The fern plants are perennial plants. Some are epiphytes while some are found on the ground.

Kingdom - Plantae

Division - Filicinophyta

Class - Pteropsida

Example: *Dryopteris*(commonly called wood fern), *Platycerium*(staghorn fern) and *Nephrolepis*

Habitat: trunks of trees or palm trees, Moist, or damp terrestrial areas e.g. moist soil.

Adaptive features of epiphytic fern to its habitat

• Sori on lower surface of frond hidden from rain.

• Shiny cuticle on upper surface of frond reduces transpiration.

• Adventitious roots along stem cling to support.

External features of a Fern

• Ferns have an underground stem/rhizome;

• With thin branched adventitious roots arising from the stem;

• Leaves/fronds develop from buds on the rhizome;

• Each frond is made of several pinnules/leaflet

• At the base of the curled leaves are the older ones which have been reduced to brown scales;

• sporangia/sori develop on the under surface of the mature leaflets;

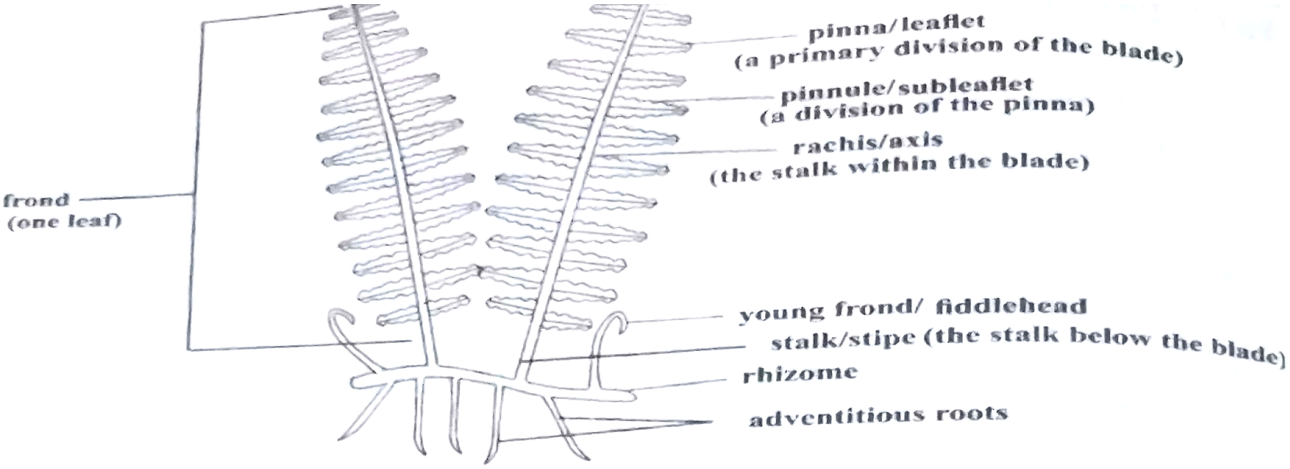
• indusium/protective layer covers each sorus.

• Each mature sporangium has a flat capsule which contains spores. The sporangium is attached to the placenta by a stalk.

• A row of cells with partially thickened walls extends three –quarter of the way around the capsule.

• These cells make up the **annulus**. The remaining one-quarter of the way around the capsule, and continuous with the annulus, is a row of large thin-walled cells which make up the **stomium**.

• The annulus and stomium provide a mechanism for throwing out the ripe spores into the air.



Drawing of the fern

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Functions of parts

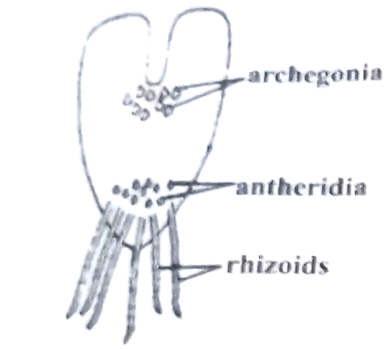
Sori made up of sporangia - spores for reproduction

Rachis - supports pinnule; photosynthesis

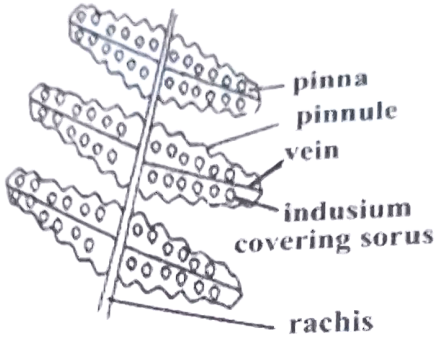
Veins - transports water, mineral salts, product of photosynthesis

Pinnule- photosynthesis

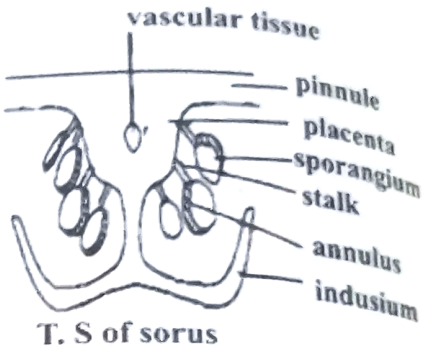
Pinna - reduces transpiration/ water loss



Underside of prothallus



Underside of leaf



T. S of sorus

Reproduction: life cycle of fern

• The life cycle of a fern consists of two alternating generations namely, the sporophyte and the gametophyte.

• The sporophyte is dominant and diploid (2n) in nature. It produces spores in sporangia borne on the leaves. The spores are haploid.

• The spores germinate into prothallus which constitutes the gametophyte.

• The gametophyte is an independent haploid plant and bears both male and female sex organs (antheridia and archegonia).

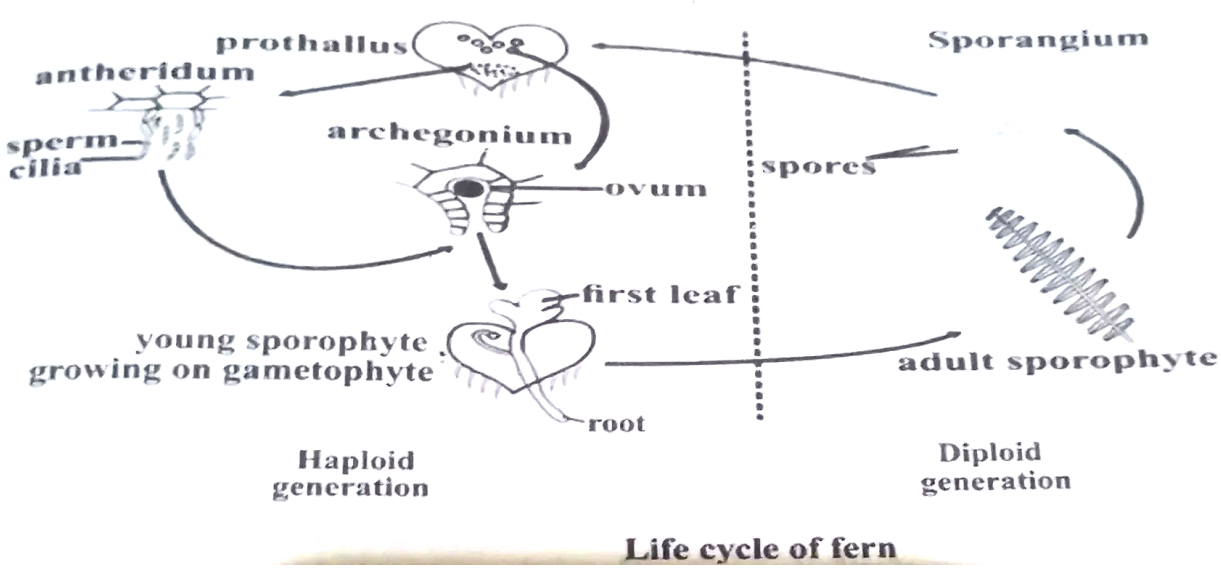
• The antheridium produces many antherizoids which are ciliated.

The archegonium produces only one ovum (or egg).

• The antherizoids swim and enter the archegonium to fuse with the ovum to form a zygote which remains in archegonium and develops into a new sporophyte.

• The sporophyte is initially dependent on the prothallus for nourishment but becomes independent as the prothallus decays.

Like mosses, ferns show **alternation of generation**. It must be noted that the gametophyte of the fern plant is small and insignificant compared to the sporophyte. In the fern plant, the sporophyte generation is the **dominant generation**.



Life cycle of fern

Haploid generation

Diploid generation

Life cycle of fern

Adaptive features fern

• They have vascular tissues for absorption of water

• The prothallus (gametophyte) has unicellular rhizoids for anchorage and absorption of water and minerals

• Fronds are large and green for maximum absorption of sunlight

• The leave surfaces are covered with cuticle to reduce excessive loss of water

• The indusium protects the sori from damage

• The sori are under the surface of the frond to prevent damage and reduce desiccation

• The antheridia and archegonia protect the gametes within them.

Features of Biological importance of specimen fern which makes it survive as an epiphyte

• Thick cuticle of the leaf / tiny brown scales on the leaf bases / tiny hairs on the lower surface of the leaf for protection.

• Rhizome / adventitious roots that can grow in the angle of the host tree for support.

• Green leaves / fronds that can photosynthesize to make food for the epiphyte.

Differences between prothallus and sporophyte

SN	Prothallus	Sporophyte
1	Temporal	Permanent
2	Rhizoids are unicellular	Roots are multicellular
3	Sex organs present	Sex organs absent
4	Spores absent	Spores present

Differences between moss and fern

1	Moss	Fern
2	Sporophyte is a parasite on the gametophyte	Sporophyte is independent on the gametophyte
3	Lacks conducting tissue	Has conducting tissue
4	Lacks true roots stems and leaves	Has true roots, stem and leaves
5	Main plant is gametophyte	Main plant is sporophyte

Differences between spore andgamete of moss

SN	Spore	Gamete
1	Develop into gametophyte formed by mitosis	Develop into sporophyte formed by meiosis
2	Formed in somatic cell	Formed in germ cell

Similarities between spore and gamete of moss

Both are haploid

Differences between Sporophyte and Gametophyte

SN	Characters	Sporophyte	Gametophyte
1	Definition	Takes part in asexual reproduction. Meiosphores do not take part in sexual reproduction, however they form gametophyte.	Multicellular, diploid structure in the life cycle of Multicellular, haploid structure in the life cyclea plant.
2	Ploidy	Diploid	Haploid
3	Formation	From zygote.	Due to germination of a meiospore: a haploid spore resulting from meiosis
4	Production	Produces spores after meiosis.	Produces gametes without meiosis.
5	Reproduction	Takes part in asexual reproduction. Meiospores do not take part in sexual reproduction, however they form gametophyte.	Takes part in sexual reproduction. Gametes fuse to form diploid zygote.

NB: 1. ***Ploidy***: The number of sets of chromosomes in a cell or an organism. For example, haploid means one set and diploid means two sets.

A meiospore of Funaria germinates into a filamentous green alga-like structure called the primary protonema

SN	Character	Antheridium	Archegonium
1	Sex organ	Male	Female
2	Shape	Rounded or club shaped in outline.	Flask shaped in outline.
3	Structure	Made up of jacket and mass of androcytes.	Made up of neck and venter.
4	Number of gametes	Produces large number of male gametes or spermatozoids.	Produces single female gamete i.e. egg or oosphere in venter.
5	Mobility of gametes	Male gametes are motile and flagellated	Female gamete is non-motile and non-flagellated.

NB: ***Androcytes give rise to the biflagellate antherozoid.***

Questions for Self-Assessment

What is the common name of the organism *Rhizopus*?

A. Bread mold B. Fern

C. Moss D. Mushroom

Which of the following is a characteristic of *Rhizopus*?

A. Autotrophic B. Hetcrotrophic

C. Monocotyledonous D. Vascular

What is the primary function of the rhizoids in *Rhizopus*?

A. Absorption of nutrients

B. Photosynthesis

C. Reproduction

D. Support

Which of the following is a type of moss?

A. Bryophyte B. Fern

C. Gymnosperm D. Angiosperm

What is the primary function of the sporophyte in mosses?

A. Absorption of nutrients

B. Photosynthesis

C. Production of spores

D. Support

Which of the following is a characteristic of ferns?

A. Autotrophic B. Heterotrophic

C. Monocotyledonous D. Vascular

What is the primary function of the fronds in ferns?

A. Absorption of nutrients B. Photosynthesis

C. Production of spores D. Support

Which of the following is a type of reproduction in *Rhizopus*?

A. Sexual B. Asexual

C. Vegetative D. Sporic

What is the primary function of the sporangia in ferns?

A. Absorption of nutrients B. Photosynthesis

C. Production of spores D. Support

Which of the following is a characteristic of mosses?

A. Autotrophic B. Heterotrophic

C. Monocotyledonous D. Vascular

What is the primary function of the protonema in mosses?

A. Absorption of nutrients B. Photosynthesis

C. Production of spores D. Support

Which of the following is a type of habitat where *Rhizopus* is commonly found?

A. Desert B. Forest

C. Freshwater D. Decaying organic matter

What is the primary function of the rhizome in ferns?

A. Absorption of nutrients B. Photosynthesis

C. Production of spores D. Support

Which of the following is a characteristic of ferns?

A. Autotrophic B. Heterotrophic

C. Monocotyledonous D. Non-vascular

What is the primary function of the spores in *Rhizopus*?

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B. Photosynthesis

C. Production of new individuals

D. Suppor

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ANSWERS

1. A	**2. B**	**3. B**	**4. B**	**5. C**
6. D	**7. B**	**8. B**	**9. C**	**10. A**
11. B	**12. D**	**13. D**	**14. A**	**15. C**

QUESTIONS (Knowledge/Recall)

What type of organism is *Rhizopus*?

Name the structure that mosses use to absorb water.

State the mode of reproduction in ferns.

Give one economic importance of *Rhizopus*.

Mention one way mosses are useful in the environment.

QUESTIONS (Understanding/ Comprehension)

Explain why *Rhizopus* is classified as a fungus.

Describe how mosses differ from ferns in their method of reproduction.

Why do ferns need water for fertilization?

Explain how *Rhizopus* obtains its nutrients.

Describe how mosses help in soil formation.

QUESTIONS (Application/Analysis)

Compare the reproductive structures of mosses and ferns.

Analyze the role of *Rhizopus* in food spoilage and in food production (e.g., in bread or alcohol).

How do mosses adapt to living in damp and shady places?

Suggest reasons why ferns are considered more advanced than mosses.

Examine the ecological importance of mosses in preventing soil erosion.

QUESTIONS (Higher Order/Case Study/Evaluation)

A farmer complains that stored bread spoils quickly during the rainy season. Using your knowledge of *Rhizopus*, explain why this happens and suggest preventive measures.

Evaluate the economic importance of ferns to humans, giving both advantages and disadvantages.

Assess the role of mosses in environmental conservation programs such as afforestation and climate regulation.

A group of students argued that ferns represent a transition between simple plants like mosses and higher plants like flowering plants. Discuss this view with evidence.

Rhizopus can be both harmful and beneficial. Justify this statement with examples.